

Center index four gives the minimal Fourier-algebra amenability constant

Candidate full solution packet for arXiv:2012.14413

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Status. Full proof, likely valid, pending specialist review. No discreteness, countability, compactness, or second-countability hypothesis is imposed.

1 Source question and answer

Yemon Choi's paper [1], arXiv:2012.14413, asks in Question 6.2 which nonabelian locally compact groups attain the minimum possible value of the Fourier-algebra amenability constant. Using his lower bound and his classification of equality for the antidiagonal constant, the paper rewrites the question as follows (physical PDF page 25):

Does $[G : Z(G)] = 4$ imply $\text{AM}(A(G)) = 3/2$?

$A(G \times G)$ need not be isometric we know from Theorem 1.4 that we do have $\text{AM}(A(G)) = \text{AD}(G)$. So if there is a group G for which $\text{AM}(A(G)) > \text{AD}(G_d)$, there must be some deeper obstruction behind this which is not visible for finite groups; Theorem 1.10 shows that we can have groups that look very different from (abelian) $\times$ (finite) where $\text{AM}(A(G)) = \text{AD}(G_d)$.

An additional motivation for Question 6.1 is that a positive answer, combined with Theorem 1.7, would give a complete characterization of those locally compact non-abelian G where $\text{AM}(A(G))$ attains its minimal value. We phrase this as a separate question, which might be easier to attack directly.

QUESTION 6.2. Which non-abelian G attain the minimal value for $\text{AM}(A(G))$?

If $\text{AM}(A(G)) = 3/2$ then, since G is non-abelian, our sharp lower bound (Theorem 1.6) implies that

$$3/2 \leq \text{AD}(G_d) \leq \text{AM}(A(G)) = 3/2$$

and so by Theorem 1.7 $|G : Z(G)| = 4$. Thus Question 6.2 is really asking if the converse holds, i.e. does $|G : Z(G)| = 4$ imply $\text{AM}(A(G)) = 3/2$? (For finite groups, the answer is positive by Theorem 1.4.)

Choi and Ghandehari subsequently proved this for discrete groups [2, Theorem 1.4(iii)], but explicitly left the general locally compact extension open. The theorem below proves that extension and, together with Choi's necessary condition, answers Question 6.2 completely.

2 Notation and proof intuition

Write $A(G)$ for Eymard's Fourier algebra and $\text{AM}(A(G))$ for its Banach amenability constant. The symbol $\widehat{\otimes}_\gamma$ denotes the Banach projective tensor product, while $\widehat{\otimes}_{\text{op}}$ denotes the operator projective tensor product. The canonical map identifies $A(G)\widehat{\otimes}_{\text{op}}A(G)$ completely isometrically with $A(G \times G)$. Multiplication is diagonal restriction,

$$\Delta u(s) = u(s, s),$$

and the bimodule actions are $(a \cdot u)(s, t) = a(s)u(s, t)$ and $(u \cdot a)(s, t) = u(s, t)a(t)$.

Proof intuition

When $[G : Z(G)] = 4$, the derived subgroup is $\{e, c\}$ for a central involution c . Translation by c splits the Fourier algebra into two ℓ^1 -summands: the even summand comes from the one-dimensional representations, and the odd summand from the two-dimensional representations.

Start with a contractive *operator* approximate diagonal and average it under simultaneous translation by c in both variables. Only two tensor blocks survive: even–even and odd–odd. The diagonal equations force each block to have asymptotic norm exactly $1/2$. Passing from the operator tensor product to the ordinary Banach projective tensor product costs 1 on the even–even block (it is the tensor square of an abelian quotient) and at most 2 on the odd–odd block (Runde’s degree bound). The resulting ordinary approximate diagonal therefore has asymptotic norm

$$\frac{1}{2} + 2\frac{1}{2} = \frac{3}{2}.$$

The main technical point is proving the exact half–half split; four matrix coefficients of one two-dimensional irreducible provide the needed identity.

3 The central spectral decomposition

Lemma 1 (Structure and spectral blocks). *Let G be a locally compact nonabelian group with $[G : Z(G)] = 4$. There is a central element $c \neq e$ such that*

$$c^2 = e, \quad [G, G] = \{e, c\},$$

and every continuous irreducible unitary representation of G has degree one or two. If $\tau : A(G) \rightarrow A(G)$ is given by $\tau a(s) = a(cs)$ and

$$A_+ = \ker(\tau - I), \quad A_- = \ker(\tau + I),$$

then

$$A(G) = A_+ \oplus_1 A_-.$$

The plus block consists precisely of the one-dimensional representation fields and is isometric to $A(G/\langle c \rangle)$; the minus block consists precisely of the two-dimensional fields.

Proof. The quotient $G/Z(G)$ is the Klein four group. Commutators depend only on cosets modulo the center, so there is a unique possible nonidentity commutator c . Comparing $[x, y]$ and $[y, x]$ gives $c = c^{-1}$, and hence $[G, G] = \{e, c\}$. The degree bound is the standard center-index-four representation lemma; equivalently, one restricts to an abelian subgroup of index two. These facts, without countability assumptions, are also proved in [1, Theorem 4.6].

Since c is central, $p_{\pm} = (1 \pm \lambda(c))/2$ are central projections in $VN(G)$. The predual decomposition of the two von Neumann summands gives the asserted ℓ^1 -sum and identifies its projections with $(I \pm \tau)/2$. Every character kills $[G, G]$. Conversely, an irreducible representation on which c acts as I factors through the abelian group $G/\langle c \rangle$ and is therefore one-dimensional. Schur’s lemma and the degree bound then put every remaining irreducible in the two-dimensional minus block. Quotient pullback gives the isometric identification of the plus block with $A(G/\langle c \rangle)$. $\square$

The same central projections in each coordinate give an additive four-block decomposition of $A(G \times G)$. We write $u^{\varepsilon\delta}$ for the block on which translation by c in the first and second variables has signs $\varepsilon, \delta \in \{+, -\}$.

4 The half-half lemma

Lemma 2 (Equal mass in the two surviving blocks). *There is a contractive operator approximate diagonal (v_α) for $A(G)$ such that*

$$v_\alpha = v_\alpha^{++} + v_\alpha^{--}, \quad \|v_\alpha^{++}\| \longrightarrow \frac{1}{2}, \quad \|v_\alpha^{--}\| \longrightarrow \frac{1}{2}.$$

Here the norms are the $A(G \times G)$, equivalently operator-projective, norms.

Proof. The group is virtually abelian and hence amenable. Ruan's operator amenability theorem supplies a contractive operator approximate diagonal (u_α) ; see also the proof of [4, Theorem 2.7]. Let $S = \tau \otimes I$, $U = I \otimes \tau$, and $T = SU$. Put

$$v_\alpha = \frac{1}{2}(u_\alpha + Tu_\alpha).$$

The map T is an isometry and $T^2 = I$. Moreover,

$$a \cdot Tu_\alpha - Tu_\alpha \cdot a = T((\tau a) \cdot u_\alpha - u_\alpha \cdot (\tau a)),$$

while $\Delta(Tu_\alpha) = \tau(\Delta u_\alpha)$. Since translating a bounded approximate identity by the involutive algebra automorphism τ gives another bounded approximate identity, (v_α) is still a contractive operator approximate diagonal. Its T -invariance kills the mixed blocks.

Set

$$e_\alpha = \Delta v_\alpha, \quad q_\alpha = \Delta(Sv_\alpha).$$

We claim that

$$bq_\alpha \longrightarrow 0 \quad \text{in } A(G) \quad (b \in A(G)). \quad (1)$$

Indeed, translate the first variable in the module-commutator defect and then restrict to the diagonal. Both operations are contractive, and the result is

$$(\tau a - a)q_\alpha \longrightarrow 0 \quad (a \in A(G)). \quad (2)$$

If $b \in A_-$, taking $a = b$ proves (1).

Now let $b \in A_+$. Continuous irreducible representations separate the points of G , so there is an irreducible π with $\pi(c) \neq I$. By Lemma 1, π is two-dimensional and $\pi(c) = -I$. In an orthonormal basis write its matrix coefficients as u_{ij} , $1 \leq i, j \leq 2$. Each u_{ij} is an odd element of $B(G)$ of norm at most one, and $bu_{ij} \in A_-$. Apply (2) to $a = bu_{ij}$ to obtain

$$bu_{ij}q_\alpha \longrightarrow 0.$$

Multiplication by $\overline{u_{ij}} \in B(G)$ is contractive on $A(G)$, and unitarity of $\pi(s)$ gives

$$\sum_{i,j=1}^2 \overline{u_{ij}(s)} u_{ij}(s) = 2.$$

Summing the four convergences yields $2bq_\alpha \rightarrow 0$. Decomposing an arbitrary b into its plus and minus parts proves (1).

Since S has signs $+1$ and -1 on the two surviving blocks,

$$\Delta v_\alpha^{++} = \frac{1}{2}(e_\alpha + q_\alpha), \quad \Delta v_\alpha^{--} = \frac{1}{2}(e_\alpha - q_\alpha).$$

Thus multiplication by either expression converges on every $b \in A(G)$ to $b/2$. Fixing b of norm one and using contractivity of Δ gives

$$\liminf_\alpha \|v_\alpha^{++}\| \geq \frac{1}{2}, \quad \liminf_\alpha \|v_\alpha^{--}\| \geq \frac{1}{2}.$$

The central von Neumann algebra decomposition makes the norms additive, while $\|v_\alpha\| \leq 1$. Hence both displayed norms converge to $1/2$. $\square$

5 Full solution

Theorem 3. *For every locally compact nonabelian group G ,*

$$\mathrm{AM}(\mathrm{A}(G)) = \frac{3}{2} \iff [G : Z(G)] = 4.$$

Proof. The forward implication is [1, Theorems 1.6 and 1.7]: the antidiagonal constant satisfies $3/2 \leq \mathrm{AD}(G_d) \leq \mathrm{AM}(\mathrm{A}(G))$, and equality in the first inequality is equivalent to center index four.

For the converse, assume $[G : Z(G)] = 4$ and take (v_α) from Lemma 2. Let

$$J : \mathrm{A}(G) \widehat{\otimes}_\gamma \mathrm{A}(G) \longrightarrow \mathrm{A}(G \times G)$$

be the canonical map. Since G is virtually abelian, J is bijective and its inverse has norm at most the maximal irreducible degree, which here is two [4, Corollary 2.6 and Theorem 2.7].

The map J respects the central block projections. On the $++$ block, J^{-1} is isometric: by Lemma 1, this is exactly the canonical Banach tensor map for the locally compact abelian group $G/\langle c \rangle$. On the $--$ block the global bound gives norm at most two. Therefore

$$\|J^{-1}v_\alpha\|_\gamma \leq \|v_\alpha^{++}\| + 2\|v_\alpha^{--}\| \longrightarrow \frac{3}{2}.$$

Because J^{-1} is an $\mathrm{A}(G)$ -bimodule map and intertwines multiplication, $(J^{-1}v_\alpha)$ is an ordinary Banach approximate diagonal. Consequently $\mathrm{AM}(\mathrm{A}(G)) \leq 3/2$. Choi's universal nonabelian lower bound gives equality. $\square$

6 Verification, scope, and novelty

The proof uses no separability hypothesis and no Plancherel measure. This is why it applies to nondiscrete groups even though the 2026 discrete proof does not. It settles Question 6.2, but it does not prove the broader conjecture $\mathrm{AM}(\mathrm{A}(G)) = \mathrm{AD}(G_d)$ for arbitrary locally compact groups, nor the full locally compact version of the sharper upper bound from [2].

The formal audit checked: preservation of the operator diagonal under simultaneous translation; the $\mathrm{A}(G)$ -norm convergence in the matrix-coefficient argument; additivity of the central predual blocks; and the exact identification of the abelian $++$ block. The bounded novelty search through 13 August 2026 found only the discrete theorem in arXiv:2507.05243, whose concluding section still presents the locally compact extension as open. Novelty is plausible, not certified.

Human-review recommendation. High priority. Check first the passage from (2) to the even case of (1), and then the isometric $++$ tensor identification. These are the two nonstandard steps; the remaining ingredients are standard Fourier-algebra and operator-amenability facts.

References

- [1] Y. Choi, *An explicit minorant for the amenability constant of the Fourier algebra*, Int. Math. Res. Not. IMRN (2023), no. 22, 19390–19430; arXiv:2012.14413.
- [2] Y. Choi and M. Ghandehari, *On amenability constants of Fourier algebras: new bounds and new examples*, J. London Math. Soc. (2026); arXiv:2507.05243.
- [3] Z.-J. Ruan, *The operator amenability of $\mathrm{A}(G)$* , Amer. J. Math. 117 (1995), 1449–1474.
- [4] V. Runde, *The amenability constant of the Fourier algebra*, Proc. Amer. Math. Soc. 134 (2006), 1473–1481; arXiv:math/0409454.